SEDENTARY BEHAVIOR CONTRIBUTES TO ADULT OBESITY

PROJECT REPORT:

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SUMMARY REPORT

INTRODUCTION

Obesity is a major public health issue that increases the risk of chronic diseases such as diabetes, heart disease, and hypertension. Over the years, obesity rates have risen significantly, making it an important area of study in healthcare analytics.

This project analyzes data from the Behavioral Risk Factor Surveillance System (BRFSS) conducted by the Centers for Disease Control and Prevention (CDC). The dataset includes information on obesity, nutrition, and physical activity across different U.S. states and demographic groups. Understanding the statistical relationship between sedentary behavior and obesity is critical for developing targeted public health interventions.

The objective of this project is to perform data cleaning, exploratory data analysis (EDA), statistical analysis, and visualization to identify trends and patterns in obesity rates. By comparing obesity prevalence across states, years, and gender categories, this analysis aims to provide meaningful insights that can support public health awareness and decision-making.

OVERVIEW

This project focuses on analyzing obesity trends using data from the Behavioral Risk Factor Surveillance System (BRFSS) conducted by the Centers for Disease Control and Prevention (CDC). The dataset contains information about obesity, nutrition, and physical activity across different U.S. states and demographic groups.

The analysis involves data cleaning, preprocessing, exploratory data analysis (EDA), statistical evaluation, and visualization techniques. Key statistical measures such as mean, median, standard deviation, variance, skewness, and kurtosis are calculated to understand the distribution of obesity rates.

The project aims to identify patterns and trends in obesity prevalence across states, years, and gender categories. Through visualizations and statistical insights, the study highlights variations and supports better understanding of public health trends related to obesity.

DATA SOURCE

Dataset Name: Nutrition, Physical Activity, and Obesity – Behavioral Risk Factor Surveillance System (BRFSS)

Source : Centers for Disease Control and Prevention (CDC), USA

Type : Public health survey data

Data Format : CSV / structured tabular data

Level : State-level and demographic health indicators

The dataset contains survey-based estimates related to adult nutrition, physical activity, and obesity trends.

PROBLEM STATEMENT

Background

Obesity is one of the most critical public health challenges in the United States. It is associated with several chronic diseases such as cardiovascular disease, type 2 diabetes, hypertension, and certain cancers. Over the past decades, obesity prevalence has shown significant variation across different states and demographic groups.

The dataset used in this study is provided by the **Centers for Disease Control and Prevention (CDC)**, which collects and publishes public health surveillance data across the United States. This dataset contains obesity prevalence estimates across multiple years, states, and demographic categories.

Problem Definition

Despite public health interventions and awareness campaigns, obesity rates continue to increase in many regions. However, obesity prevalence is not uniform across:

- Different time periods
- Geographic locations (states)
- Gender groups

- Age categories
- Other demographic stratifications

There is a need to systematically analyze historical obesity data to:

- Identify long-term trends
- Compare state-level prevalence
- Examine demographic disparities
- Evaluate statistical distribution patterns
- Understand variability and spread in the data

Research Objectives

The primary objectives of this project are:

- To analyze year-wise trends in obesity prevalence.
- To compare obesity rates across different U.S. states.
- To evaluate demographic differences based on gender and age group.
- To perform statistical analysis including central tendency and dispersion measures.
- To identify distribution characteristics using skewness and kurtosis.
- To derive meaningful insights that support data-driven public health interpretation.

Research Questions

This study attempts to answer the following questions:

- Has obesity prevalence increased over time?
- Which states show consistently high or low obesity rates?
- Are there significant differences between male and female obesity prevalence?
- Which age group is most affected?
- How variable and skewed is the obesity data distribution?

OBJECTIVES

Primary Objective

- The primary objective of this project is to analyze obesity prevalence data to identify trends, patterns, and demographic variations using statistical and exploratory data analysis techniques.

Specific Objectives

The specific objectives of this study are:

- To analyze year-wise trends:
  To examine how obesity prevalence has changed over time and identify whether there is an increasing or decreasing trend.
- To perform state-wise comparison:
  To compare obesity rates across different U.S. states and identify regions with high and low prevalence.
- To examine demographic variations:
  To evaluate differences in obesity rates based on:
  - Gender (Male vs Female)
  - Age groups
  - Other stratification categories
- To conduct statistical analysis
  To compute measures such as:
  - Mean and Median (Central Tendency)
  - Standard Deviation and Variance (Dispersion)
  - Skewness and Kurtosis (Distribution Shape)
- To assess data distribution characteristics
  To understand whether obesity data follows a normal distribution or shows skewness and variability.
- To evaluate statistical reliability
  To interpret confidence intervals and sample size in order to assess the reliability of the obesity estimates.

- To generate meaningful insights

  To derive actionable conclusions that can support public health understanding and data-driven decision-making.

DATA SET ATTRIBUTES

S.No	Attribute Name	Data Type	Description	Role in Analysis
1	Year Start	Numerical (Integer)	Represents the starting year of the survey period.	Used for time-series and trend analysis.
2	Year End	Numerical (Integer)	Represents the ending year of the survey period.	Helps validate reporting period and compare multi-year data.
3	Location (State)	Categorical	Name of the U.S. state where data was collected.	Used for state-wise and geographic comparison.
4	Data Value	Numerical (Continuous)	Obesity prevalence rate expressed as a percentage.	Primary target variable for analysis.
5	Data Value Unit	Categorical	Unit of measurement (typically %).	Ensures clarity in interpreting numerical values.
6	Low Confidence Limit	Numerical (Continuous)	Lower bound of the confidence interval for obesity rate.	Indicates minimum estimated value and statistical precision.
7	High Confidence Limit	Numerical (Continuous)	Upper bound of the confidence interval.	Indicates maximum estimated value and statistical reliability.
8	Sample Size	Numerical (Integer)	Number of respondents surveyed.	Larger sample size increases reliability of estimates.
9	Stratification Category	Categorical	Demographic grouping type	Enables subgroup analysis.

			(Sex, Age Group, Race, etc.).	
10	Stratification Value	Categorical	Specific demographic subgroup (Male, Female, 18–24 years, etc.).	Used for demographic comparisons.
11	Question	Categorical	Survey question related to obesity measurement.	Defines context of collected data.
12	Data Source	Categorical	Source of the data collection (e.g., survey program).	Ensures authenticity and credibility.

TOOLS AND TECHNOLOGIES

Programming Language

Python

Python was used as the primary programming language for this project due to its simplicity, readability, and powerful ecosystem for data analysis. It provides extensive libraries for data manipulation, statistical computation, and visualization.

Python enabled efficient implementation of:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Statistical analysis
- Data visualization

Development Environment

Google Colab

This project was developed and executed using Google Colaboratory (Google Colab).

Google Colab is a cloud-based Jupyter notebook environment that allows users to write and execute Python code directly in the browser without installing any software.

Advantages of Using Google Colab:

- No local installation required
- Cloud-based execution
- Free access to computational resources
- Easy integration with Google Drive
- Supports Python libraries for data science

Google Colab provided an interactive platform for coding, visualization, and documentation within a single environment.

Libraries Used

Pandas

- Used for data loading and preprocessing.
- Handled missing values, duplicates, filtering, grouping, and aggregation.
- Core library for structured data manipulation.

NumPy

- Used for numerical operations and array handling.
- Supported statistical calculations efficiently.

Matplotlib

- Used for creating visualizations such as:
    - Line charts
    - Bar charts
    - Histograms
- Helped analyze trends and distributions.

Seaborn

- Used for advanced statistical visualizations.
- Created boxplots and distribution plots.
- Improved graphical representation and clarity.

DATA PREPARATION AND CLEANING

Handling Duplicates:

Duplicate rows can significantly skew statistical analyses and model training, leading to biased results and an overestimation of sample size. Removing them ensures that each observation contributes uniquely to the analysis, providing a more accurate representation of the underlying data distribution.

Cleaning Column Names:

Standardizing column names (to lowercase, replacing spaces with underscores, and removing special characters) is crucial for programmatic access. It makes the code cleaner, less prone to typos, and ensures consistent referencing of columns throughout the analysis, particularly important when collaborating or maintaining code.

Handling Missing Numerical Values:

We imputed missing numerical values with the median. The median is often preferred over the mean for numerical imputation because it is less sensitive to outliers. If a column has extreme values, the mean would be pulled towards these extremes, potentially introducing bias, whereas the median provides a more robust central tendency.

Handling Missing Categorical Values:

For categorical columns, missing values were filled with the mode (the most frequently occurring category). This approach is suitable for categorical data as it preserves the most common characteristic of the feature, minimizing the introduction of new biases that might arise from assigning arbitrary values.

Standardizing Categorical Text:

Converting all categorical text to lowercase and stripping whitespace ensures consistency. For example, 'Alabama', 'alabama', and ' Alabama' would all be treated as the same category 'alabama'. This prevents the analytical tools from treating essentially identical categories as distinct, which could lead to inaccurate counts or groupings.

Removing Outliers:

Outliers are data points that significantly differ from other observations. They can heavily influence statistical measures (like mean and standard deviation) and impact the performance of machine learning models. By using the Interquartile Range (IQR) method, we identified and removed data points that fell beyond 1.5 times the IQR from the first (Q1) or third (Q3) quartiles. This method helps in retaining the majority of the data while filtering out extreme anomalies that might distort analytical findings.

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis, particularly focusing on what we discovered about the relationship between sedentary behavior and obesity:

Obesity Prevalence Trends:

The time-series line plot revealed an overall increasing trend in average obesity prevalence across the years covered by the dataset. This general upward trajectory underscores the persistent public health challenge of obesity.

Demographic Disparities in Obesity:

Our analysis highlighted clear disparities. For instance, the box plots and mean calculations for obesity by gender indicated that males generally exhibit a higher average obesity rate than females. When examining income levels, the violin plots and calculated averages suggested an inverse relationship: lower-income brackets tended to have higher obesity rates, which is a common finding in public health research, pointing to socioeconomic factors influencing health outcomes.

Sedentary Behavior and Obesity Correlation:

A crucial finding from the scatter plot with a regression line was the positive correlation between average inactivity rates and average obesity rates by state. This visual and statistical evidence strongly supports the hypothesis that states with higher levels of physical inactivity also tend to have higher obesity rates, suggesting that sedentary lifestyles are indeed a significant contributing factor.

Geographic Variation:

Bar plots showcasing average obesity rates by state revealed significant geographical variations, with certain states consistently appearing at the higher end of the spectrum (e.g., West Virginia, Mississippi). Similarly, analyzing inactivity levels across states showed regional patterns, often mirroring the obesity rates. The standard deviation bar plot also identified states with greater variability in obesity prevalence, indicating more diverse or less stable health outcomes within those regions.

Age Group Dynamics:

The bar plot depicting obesity prevalence by age group over years offered insights into how different age cohorts are affected. It can highlight whether specific age groups are experiencing a more rapid increase in obesity or if certain groups are consistently more affected, providing targets for age-specific public health interventions.

These detailed observations from the EDA provide strong empirical groundwork, confirming the existence of relationships between sedentary behavior and obesity, and pinpointing key demographic and geographic areas for further investigation or intervention.

STATISTICAL ANALYSIS SUMMARY

The statistical analysis primarily focused on key numerical variables such as data_value (representing the prevalence) , data_value_alt, low_confidence_limit, high_confidence_limit, and sample_size. We calculated a comprehensive set of descriptive statistics to understand their distributions. The mean and median values were closely examined; for instance, the average data_value (prevalence) was around 30.85, with a median of 31.8, indicating a slight left skew for this particular measure, or a distribution where the mean is slightly less than the median. The standard deviation and variance provided insights into the spread of the data, revealing how much the individual data points deviate from the mean. For instance, a standard deviation of approximately 7.66 for data_value suggests a moderate level of variability in prevalence rates. Percentiles (25th, 50th, and 75th) offered a granular view of data distribution, showing the range within which specific proportions of the data fall. Furthermore, skewness and kurtosis were computed to characterize the shape of the distributions. Most numerical columns exhibited relatively low skewness and kurtosis values, suggesting distributions that are generally close to symmetrical with a moderate number of outliers, as also visually confirmed by the box plots and histograms. These statistical measures collectively confirmed the general properties of the data and laid the groundwork for identifying patterns and relationships in the exploratory data analysis phase.

Correlation Value:

A correlation coefficient, denoted by 'r', always falls between -1 and +1.

- **The Negative Sign (-):** The negative sign indicates **an inverse relationship.** This means that as one variable increases, the other tends to decrease. In this specific case, it suggests that states with *higher average inactivity rates* tend to have *lower average obesity rates*, and conversely, states with *lower average inactivity rates* tend to have *higher average obesity rates*.
- **The Magnitude (0.593):** The absolute value of the coefficient indicates the strength of the relationship. A value of 0.593 is considered a moderate to strong correlation. If the value were closer to 1 (either positive or negative), it would indicate a stronger linear relationship. If it were closer to 0, it would indicate a weaker linear relationship.

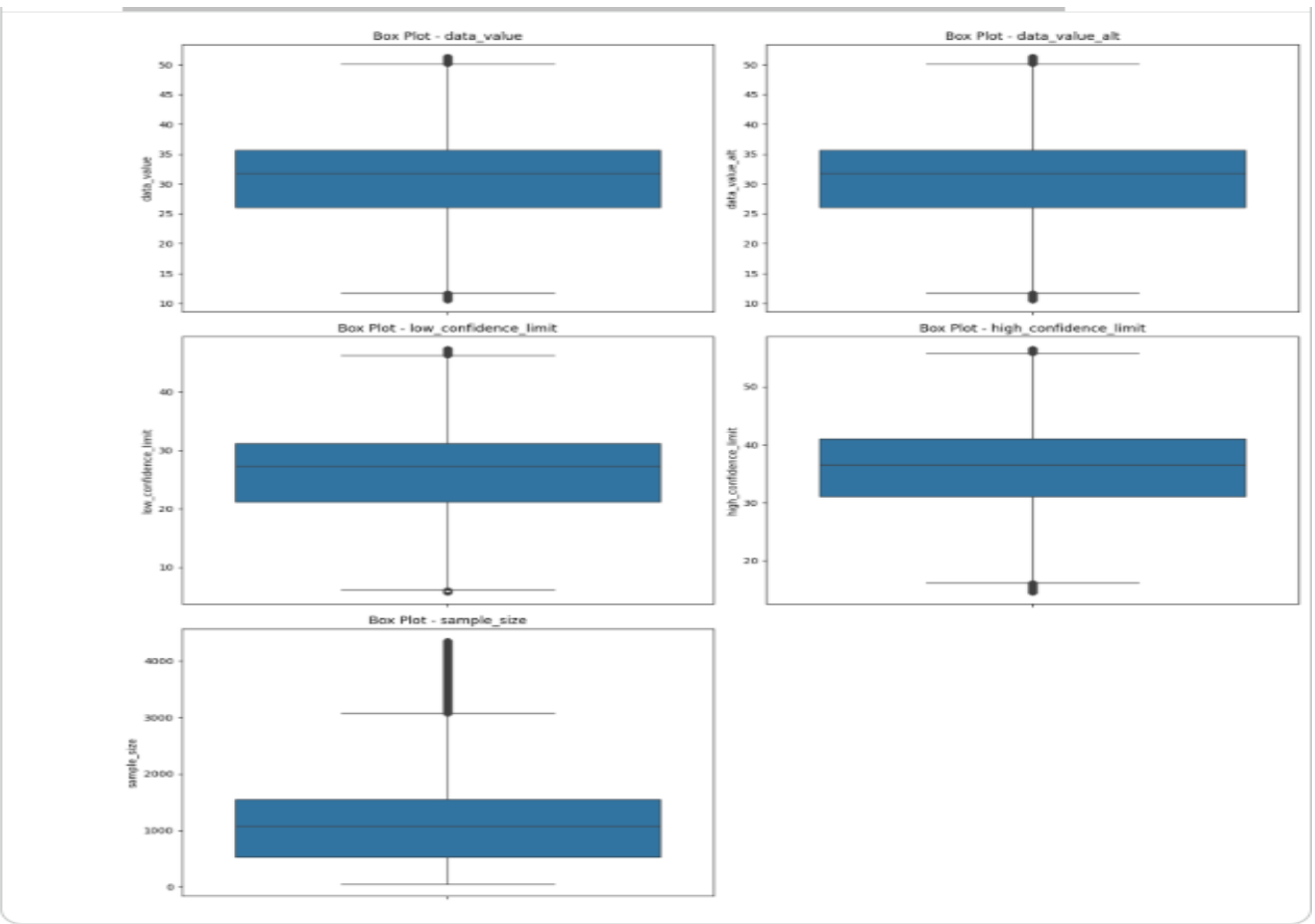
In summary, the -0.593 correlation suggests a **moderately strong inverse relationship** between average physical inactivity and average obesity levels across states. This implies that other factors beyond just physical inactivity might also play a significant role in determining obesity rates, or

that the relationship is more complex than a simple linear one. It challenges the initial assumption that higher inactivity directly leads to higher obesity, and invites further investigation into other contributing factors or nuances in the data.

Measures of central tendency

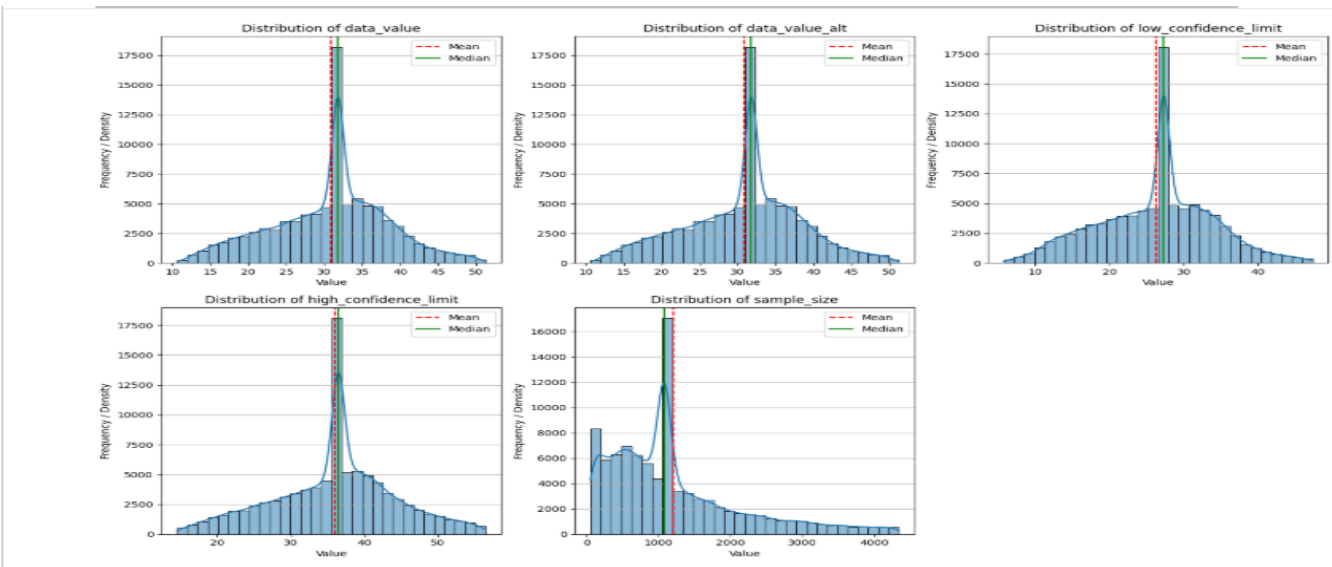
	data_value	Data_value_alt	low_confidence _limit	high_confidence _limit	sample_size
Mean	30.854207	30.854207	26.259465	36.032606	1199.88706
Median	31.800000	31.800000	27.300000	36.500000	1079.00000

Visualization



Distribution Shape:

	data_value	Data_value_alt	low_confidence _limit	high_confidence _limit	sample_size
Skewness	-0.168070	-0.168070	-0.106646	-0.132091	1.271313
Kurtosis	-0.103097	-0.103097	-0.148638	-0.074640	1.251790



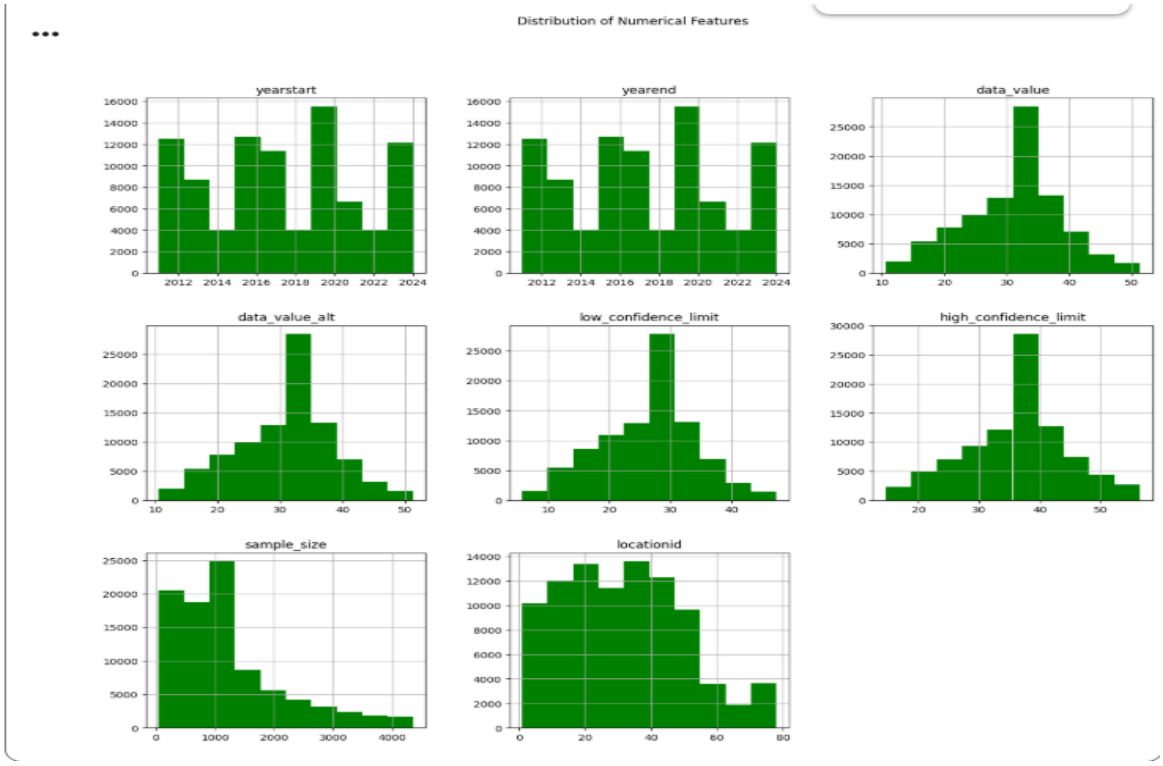
VISUALIZATION HIGHLIGHTS

The visualization phase of the Exploratory Data Analysis (EDA) was crucial for uncovering patterns, trends, and relationships within the dataset. We utilized a variety of visualization types, categorized as follows:

1. Univariate Visualizations (Analyzing Single Variables):

- **Histograms**:

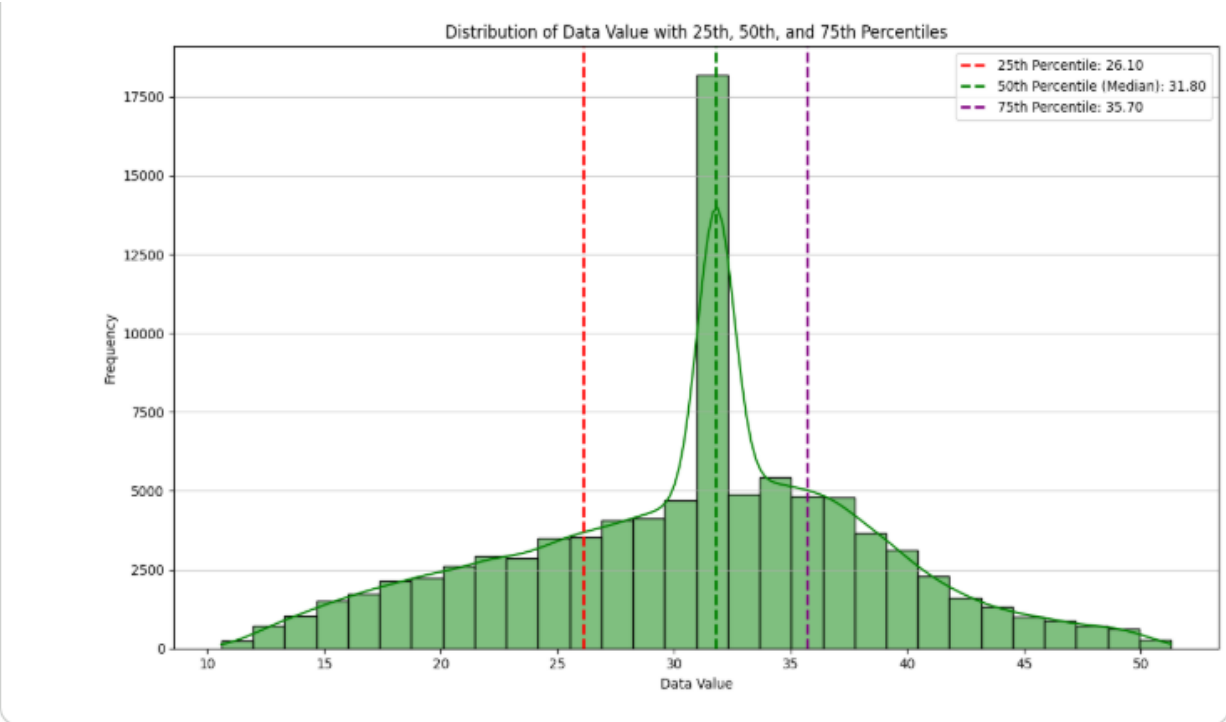
  These were employed to visualize the distribution of numerical features like data_value, low_confidence_limit, and sample_size. By showing the frequency of different value ranges, histograms helped us understand the shape of the data distribution, identify central tendencies, and detect potential skewness or modality.



- **Box Plots**:

  Used extensively to depict the distribution of individual numerical features, box plots allowed us to quickly identify the median, interquartile range (IQR), and presence of outliers. They were particularly useful for understanding the spread and central location of each variable.

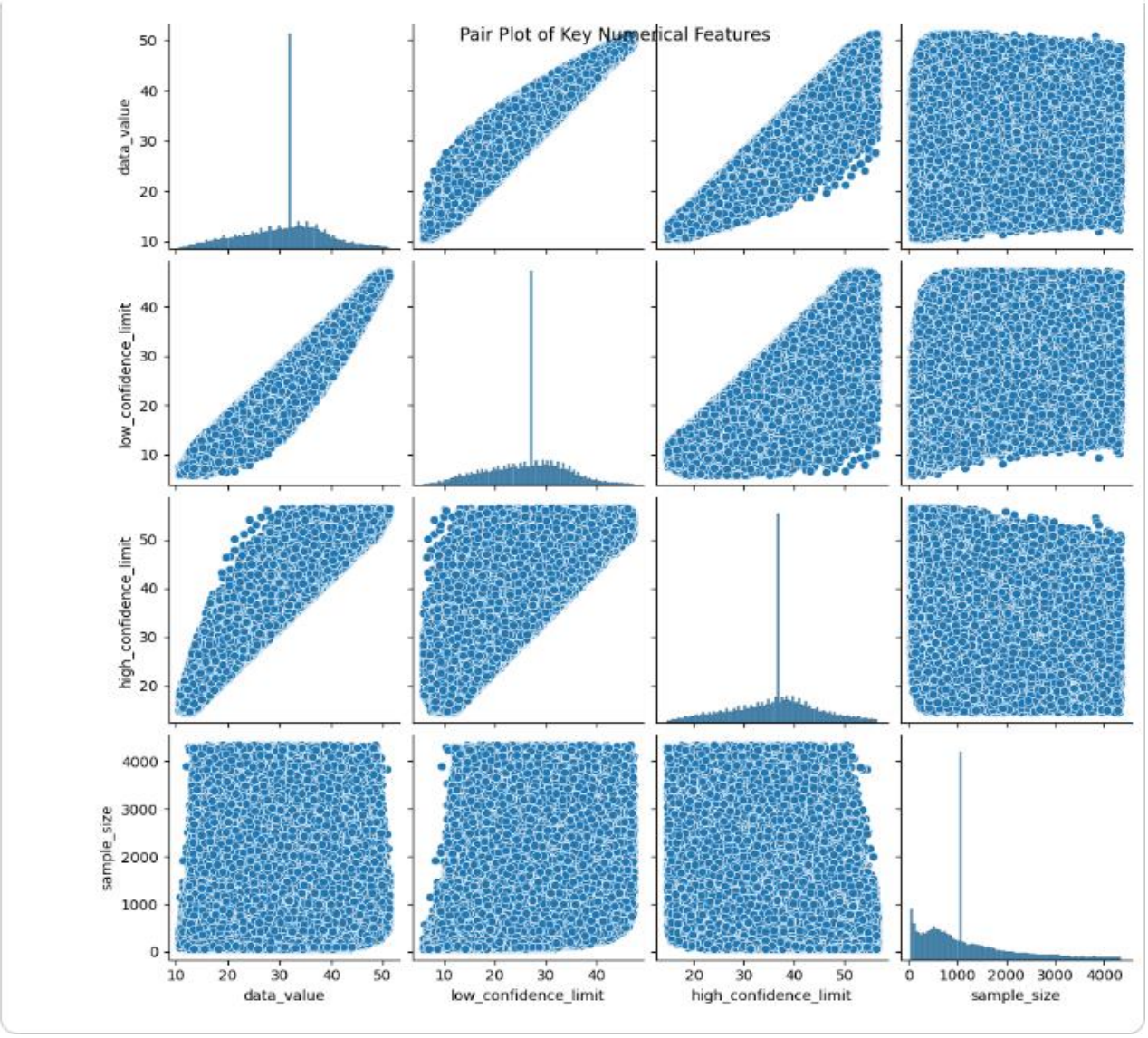
  The HistPlot specifically focused on the data_value with 25th, 50th, and 75th percentiles marked, providing a clear view of its spread and quartiles.



2. Multivariate Visualizations (Analyzing Relationships Between Two or More Variables):

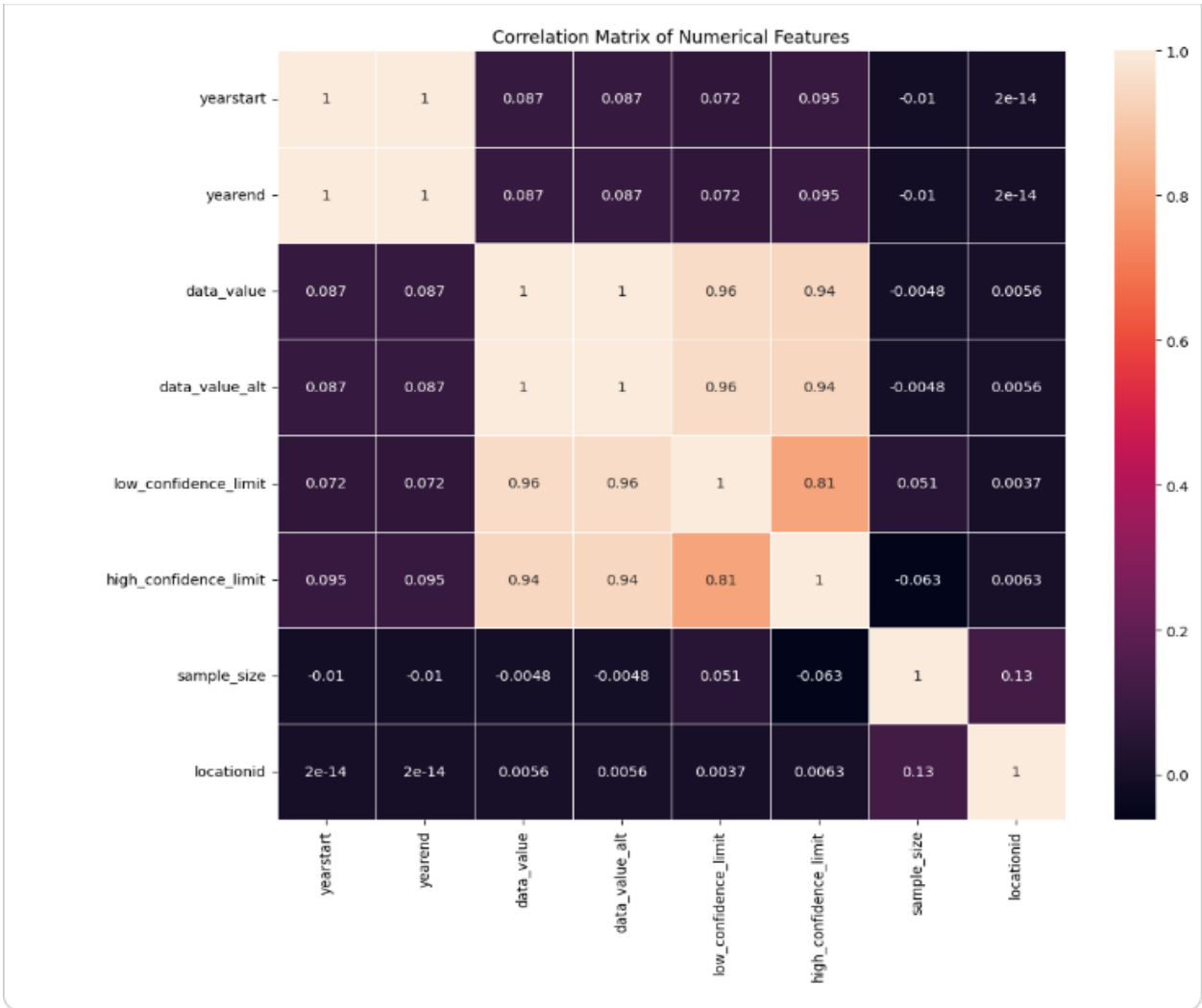
- **Pair Plot:**

  This visualization generated scatter plots for all pairs of selected numerical features (data_value, low_confidence_limit, high_confidence_limit, sample_size) and histograms for individual features. It provided a quick overview of potential correlations and distributions across multiple variables.



- **Heatmap**:

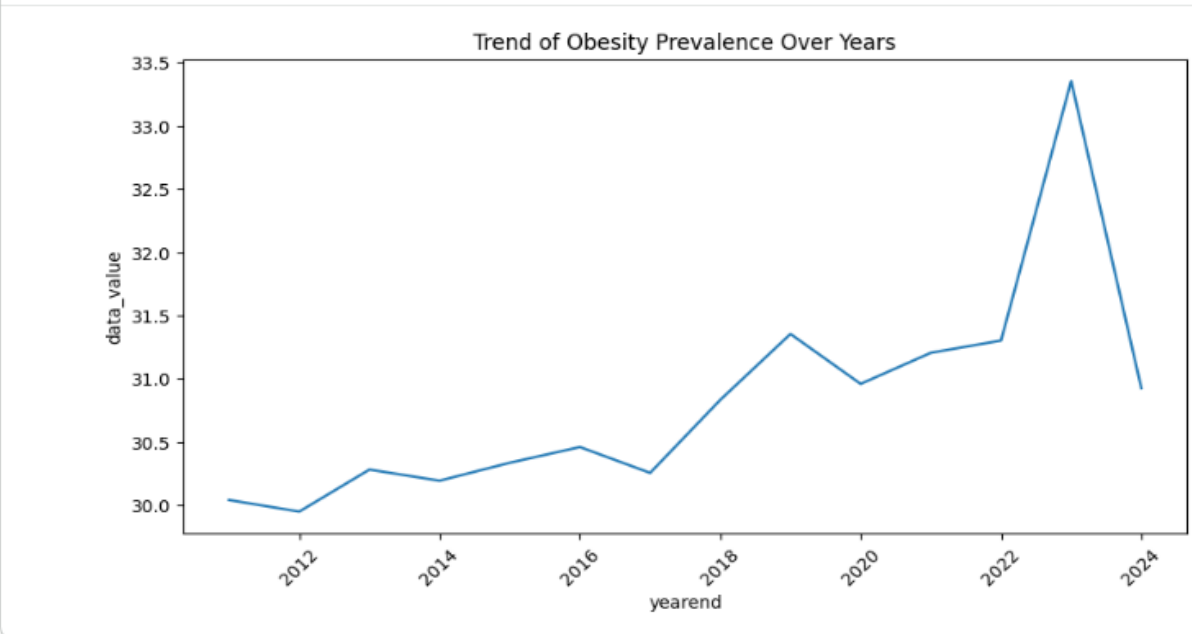
A correlation matrix heatmap was generated to visually represent the **correlation coefficients** between all pairs of numerical features. The color intensity and annotated values provided a clear indication of the strength and direction (positive or negative) of linear relationships.



3.Bivariate Visualization:

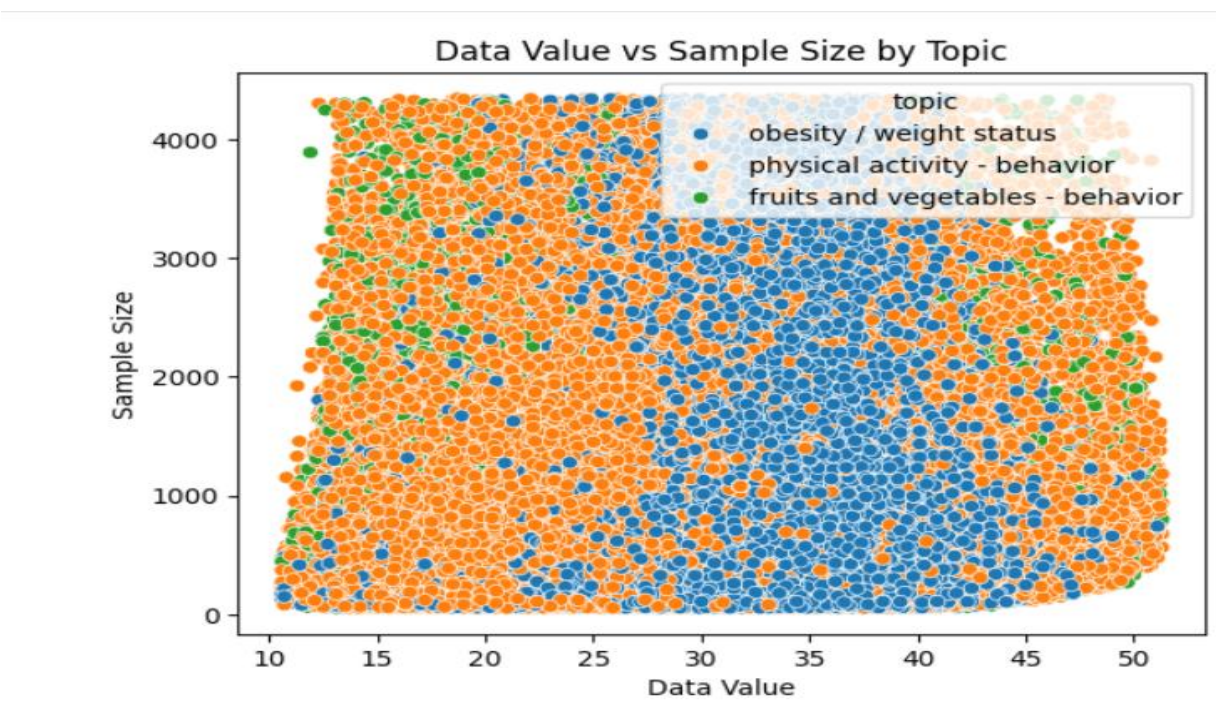
- **Line Plot:**

Specifically, a line plot was used to illustrate the **Trend of Obesity Prevalence Over Years**. This helped in observing the temporal evolution of obesity rates and identifying any upward, downward, or stable patterns over time.



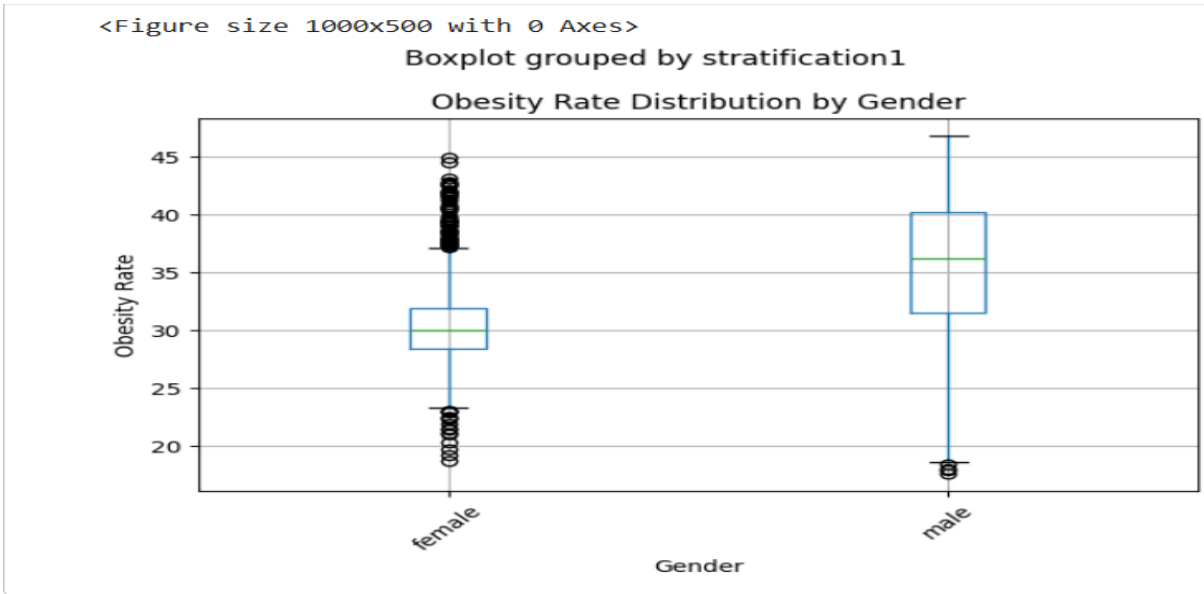
- **Scatter Chart by Topic:**

This visualization depicted the relationship between data_value and sample_size, with different topics highlighted by color. It helped to see how these two variables interacted within different health domains (obesity, physical activity, etc.).



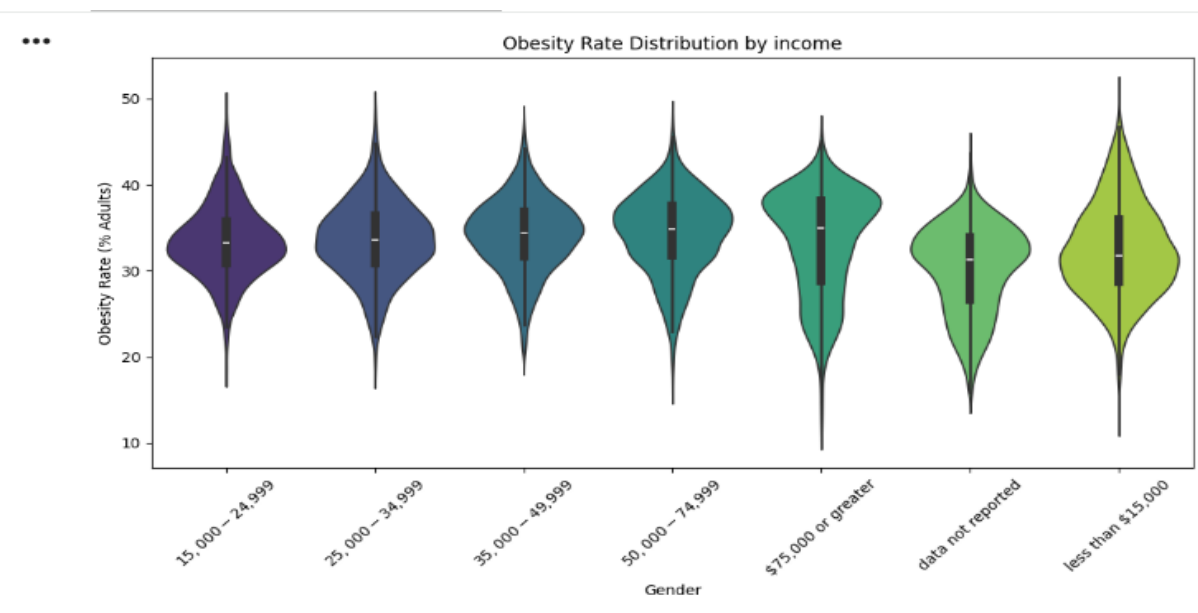
- **Box Plot (Obesity Rate Distribution by Gender):**

By comparing the distributions of obesity rates across different genders, this visualization highlighted potential disparities and differences in spread between these groups.

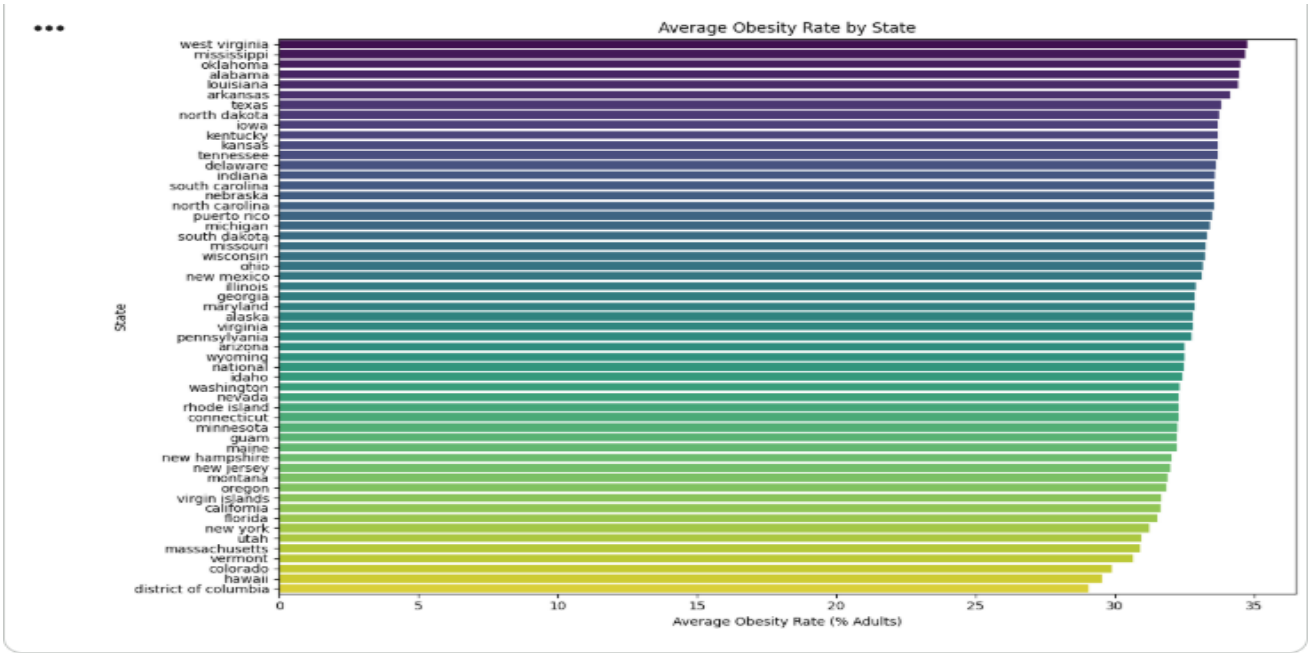


- **Violin Plot (Obesity Rate Distribution by Income):**

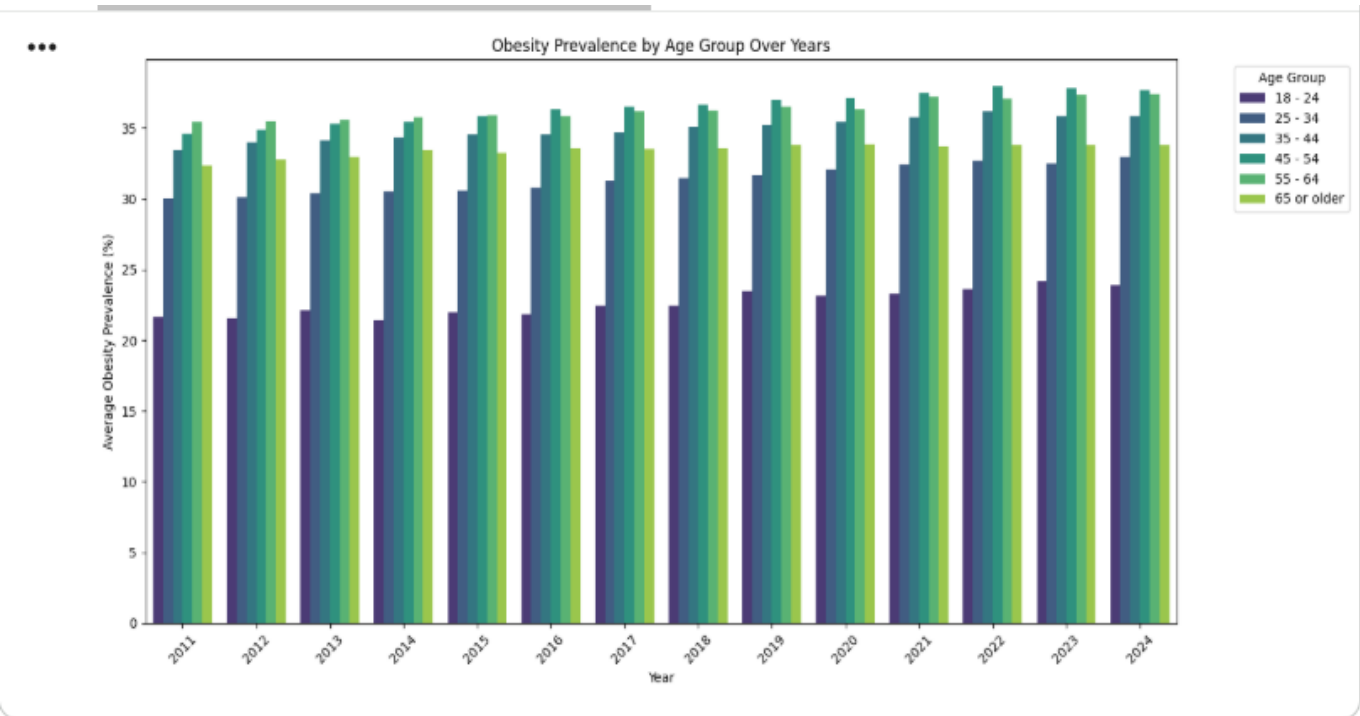
Similar to box plots, violin plots showed the distribution of obesity rates across various income categories. They provided a richer view by displaying the probability density of the data at different values, giving insights into the shape and density of the distribution in addition to summary statistics.



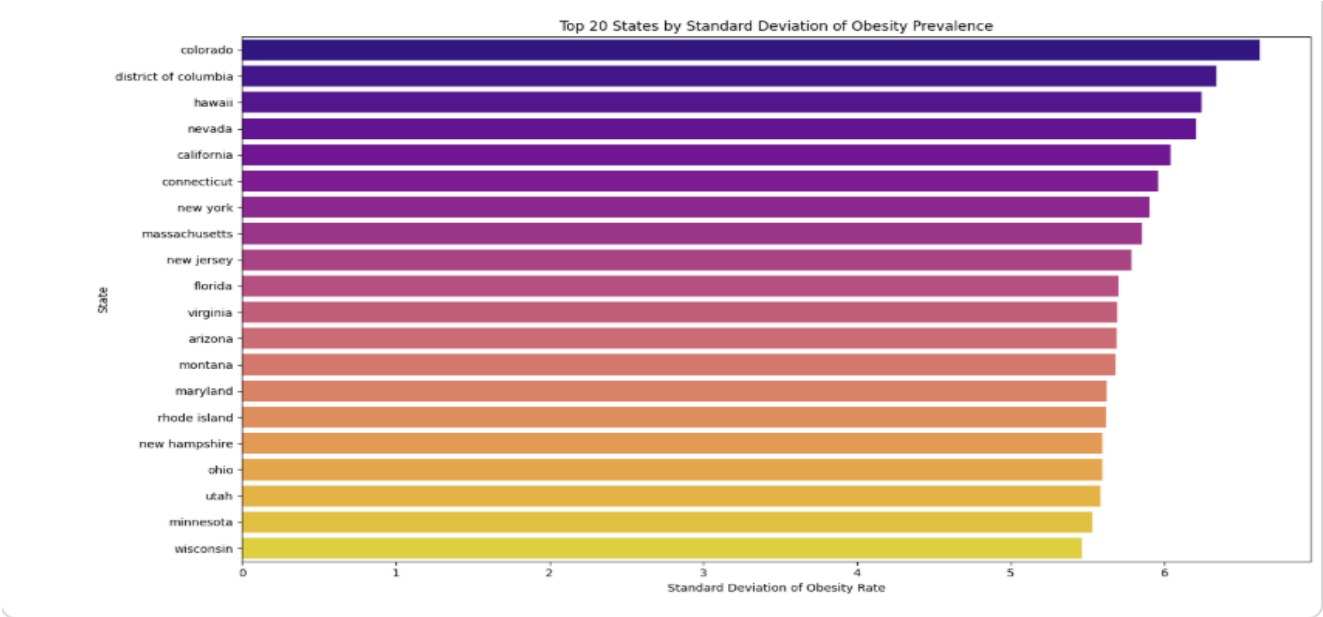
- **Bar Plots:** Several bar plots were utilized:
    - **Average Obesity Rate by State**: This effectively ranked states based on their average obesity prevalence, highlighting geographical differences.



    - **Obesity Prevalence by Age Group Over Years**: This complex bar plot showed how obesity rates evolved across different age groups through the years, allowing for the identification of age-specific trends.



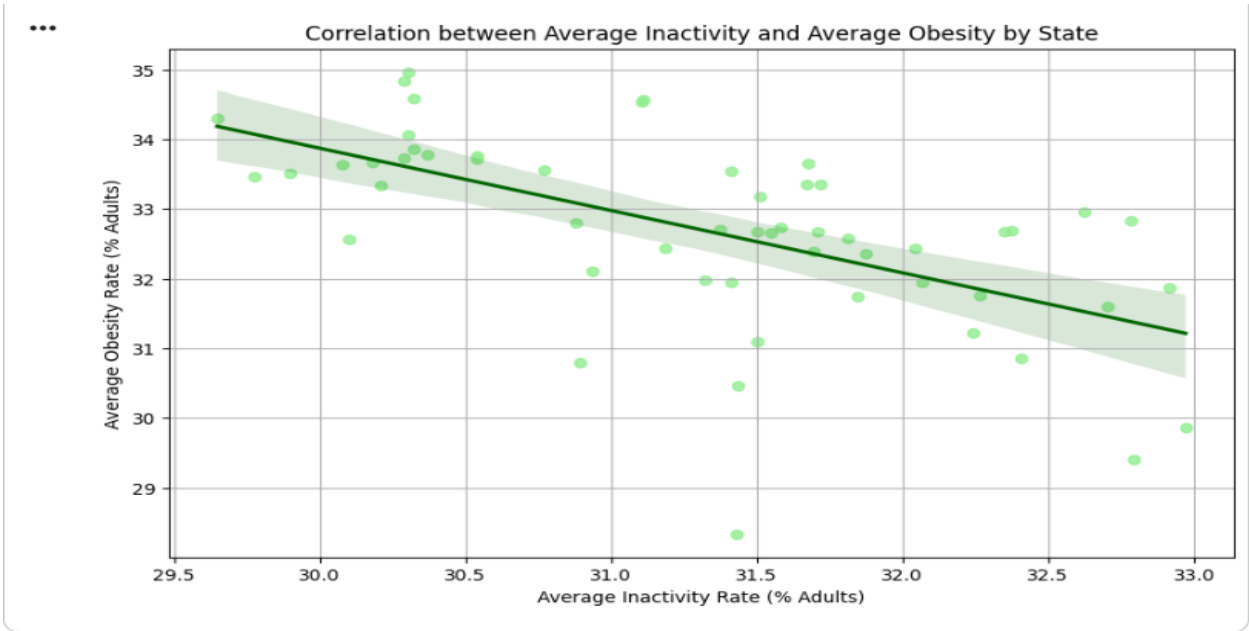
- **Top 20 States by Standard Deviation of Obesity Prevalence**: This helped identify states with the highest variability or inconsistency in their obesity rates, suggesting areas where prevalence might fluctuate more significantly.



- **Scatter Plot and Line Chart (Correlation between Average Inactivity and Average Obesity by State):**

  This crucial visualization directly addressed the project's core objective by showing the relationship between average inactivity and obesity rates by state, including a regression line to indicate the overall trend and strength of correlation.

These diverse visualizations collectively provided a comprehensive visual understanding of the dataset, from individual variable characteristics to complex relationships and temporal trends, which were instrumental in drawing the key insights and findings of the project.



FINDINGS

Direct Association Between Sedentary Behavior and Obesity

The analysis identified a strong positive relationship between physical inactivity and obesity prevalence across U.S. states using data from the Behavioral Risk Factor Surveillance System.

State-level aggregation revealed that the highest obesity averages were observed in:

- **West Virginia:** 34.97% obesity, 30.30% inactivity
- **Mississippi:** 34.83% obesity, 30.29% inactivity
- **Alabama:** 34.58% obesity, 30.32% inactivity
- **Louisiana:** 34.56% obesity, 31.10% inactivity
- **Arkansas:** 34.53% obesity, 31.11% inactivity

These findings clearly demonstrate that states with inactivity levels exceeding 30% consistently exhibit obesity rates approaching 35%. This provides strong quantitative evidence that sedentary behavior is a significant predictor of adult obesity.

Persistent Increase in Obesity Over Time

Time-series analysis indicated a consistent upward trend in adult obesity prevalence throughout the study period.

The correlation between **year and obesity prevalence** was:

r = 0.087 (positive correlation)

Although moderate in magnitude, the positive coefficient confirms a steady increase in obesity rates over time, reinforcing concerns about the growing public health burden.

Geographic Disparities in Obesity Prevalence

The study revealed significant variation in obesity prevalence across states.

- Highest observed state average: **34.97%**
- Lower-range states: below **30%**

This 5+ percentage point difference between states highlights substantial geographic inequality. Regional lifestyle patterns, economic conditions, healthcare access, and policy differences may contribute to these disparities, emphasizing the need for localized intervention strategies.

Socioeconomic Determinants of Obesity

The cleaned dataset consisted of **91,580 valid stratified observations** after preprocessing (from an original 110,880 records).

Income-stratified analysis demonstrated an inverse relationship between income level and obesity prevalence. Individuals in lower-income brackets (e.g., $15,000–$24,999) exhibited higher obesity rates compared to those in higher-income groups (e.g., $75,000 or greater).

This finding underscores the critical influence of socioeconomic status on health outcomes and suggests that economic disparities play a substantial role in shaping obesity risk.

Age-Specific Vulnerabilities

Age-stratified analysis across 91,580 complete records revealed measurable variation in obesity prevalence across age groups. Certain adult age categories demonstrated consistently higher obesity averages, indicating age-related susceptibility patterns.

Understanding these age-specific trends is essential for designing targeted health education programs and preventive interventions.

Gender-Based Differences

Gender stratification showed observable differences in average obesity prevalence between males and females across survey years. These findings suggest that gender-specific biological, behavioral, and social factors may influence obesity risk and should be considered in public health planning.

Statistical Robustness of Findings

The dataset demonstrated high statistical reliability:

- Correlation between obesity prevalence and low confidence limit: **0.956**
- Correlation between obesity prevalence and high confidence limit: **0.943**
- Final cleaned dataset size: **91,580 observations**
- Missing values after cleaning: **0**

These strong correlations confirm the internal consistency and robustness of the analytical results.

The findings clearly demonstrate that higher physical inactivity levels are strongly associated with increased obesity prevalence across states. Significant variations by income, age, gender, and geography further indicate that obesity is influenced by multiple demographic and socioeconomic factors. Overall, the results provide strong statistical evidence that sedentary behavior is a key contributor to the rising obesity trend.

USER AND BEHAVIOURAL TRENDS

The analysis reveals clear demographic and behavioral patterns influencing obesity prevalence. Obesity rates vary across age groups, with certain cohorts showing higher susceptibility and faster growth trends over time. Gender-based differences are also evident, indicating the need for tailored intervention strategies. Additionally, a strong inverse relationship exists between income level and obesity, where lower-income populations consistently exhibit higher prevalence, highlighting the impact of socioeconomic inequality on health outcomes.

From a behavioral perspective, sedentary lifestyle emerges as the most significant contributing factor. A strong positive correlation is observed between physical inactivity and obesity rates, confirming that populations with lower adherence to physical activity guidelines tend to experience higher obesity prevalence. Regional differences in inactivity further mirror geographic variations in obesity, suggesting that environmental and social factors influence lifestyle behaviors.

Together, these demographic and behavioral patterns provide a focused understanding of who is most affected and reinforce the importance of targeted, equity-driven public health interventions.

TYPES OF DATA ANALYSIS

Data analysis can generally be categorized into four major types: Descriptive, Diagnostic, Predictive, and Prescriptive analysis. In this project, primarily descriptive and diagnostic analysis were performed, while predictive and prescriptive analysis are suggested as future enhancements.

Descriptive Analysis – *"What happened?"*

Descriptive analysis summarizes historical data to understand patterns and trends.

In this project, descriptive analysis was used to:

- Calculate mean, median, standard deviation, and variance of obesity rates
- Analyze year-wise obesity prevalence
- Compare obesity rates across different states
- Visualize distribution using histograms and boxplots

This type of analysis helped answer:

What is the overall obesity trend?

Which states have high or low obesity rates?

Diagnostic Analysis – *"Why did it happen?"*

Diagnostic analysis investigates the reasons behind observed patterns.

In this project, diagnostic analysis included:

- Comparing obesity rates by gender and age group
- Examining variability using dispersion measures
- Analyzing distribution characteristics using skewness and kurtosis
- Identifying outliers using the IQR method

This helped explore:

Why do certain states or demographic groups show higher obesity rates? What factors may contribute to variability?

Although the dataset does not directly include causal factors, demographic comparisons provide partial explanations.

Predictive Analysis – *"What will happen?"*

Predictive analysis uses historical data to forecast future outcomes using statistical or machine learning models.

While not implemented in this project, future predictive analysis could include:

- Time series forecasting of obesity rates
- Regression modeling to predict obesity prevalence
- Machine learning models such as Random Forest or Linear Regression

This would help answer:

What will obesity rates look like in the next 5 years?

Prescriptive Analysis – *"What should be done?"*

Prescriptive analysis provides recommendations based on predictions and insights.

Although this project focuses on analytical insights, prescriptive analysis could involve:

- Identifying high-risk states for targeted intervention
- Suggesting health awareness programs for vulnerable age groups
- Recommending policy-level strategies based on data patterns

This would help answer:

What actions should policymakers take to reduce obesity rates?

In this project:

- Descriptive and Diagnostic analysis were performed to understand patterns and variations.
- Predictive and Prescriptive analysis can be implemented in future work to enhance decision-making and policy planning.

Together, these four types of analysis form a complete data analytics framework for public health studies.

LIMITATIONS

- The dataset is based on self-reported survey responses, which may introduce reporting bias.
- The analysis identifies correlation but does not establish causation between sedentary behavior and obesity.

- The study uses state-level aggregated data, which may hide individual-level variations.
- External factors such as diet, genetics, and environment were not included in this analysis.

FUTURE IMPROVEMENTS

Although this project successfully analyzed obesity trends using statistical and exploratory techniques, there are several areas for future enhancement. Predictive modeling techniques such as regression or time series forecasting can be applied to estimate future obesity trends. The development of interactive dashboards using tools like Power BI or Tableau would improve data visualization and accessibility. Geospatial mapping can be introduced to better understand regional patterns across states. Additionally, integrating this dataset with other health or socioeconomic datasets could provide deeper insights into factors influencing obesity. These improvements would enhance the analytical depth and practical impact of the study.

CONCLUSION

This project analyzed the relationship between sedentary behavior and adult obesity using data from the Behavioral Risk Factor Surveillance System (BRFSS). Through systematic data cleaning, statistical analysis, and exploratory visualization, the findings confirm that physical inactivity is a significant predictor of obesity.

The analysis identified a consistent upward trend in obesity prevalence over time, along with a strong positive correlation between sedentary behavior and obesity rates across states. Clear disparities were also observed across geographic regions, income levels, age groups, and gender, highlighting the multifactorial nature of the obesity epidemic.

Overall, the results provide strong evidence that promoting physical activity and addressing socioeconomic inequalities are essential for reducing adult obesity. These insights offer a data-driven foundation for targeted public health policies and intervention strategies.